



PRODUCT DATA SHEET

GROWTH & GH AXIS

RESEARCH USE ONLY

GROWTH & GH AXIS

Tesamorelin

Tesamorelin – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Tesamorelin
SYNONYMS / ALIASES	TH9507; trans-3-hexenoyl-GHRH(1-44) amide
CAS NUMBER	218949-48-5
MOLECULAR FORMULA	C221H366N72O67S
MOLECULAR WEIGHT	Approximately 5135.9 g/mol
AMINO ACID SEQUENCE	44-amino-acid analog of human GHRH (YADAIFTNSYRKVLGQLSARKLLQDIMSRQQGESNQERGARARL with C-terminal amide), modified at the N-terminus by a trans-3-hexenoyl group
SOURCE	Synthetic (solid-phase peptide synthesis with N-terminal acylation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 7-14 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Contains a methionine residue susceptible to oxidation; protect from prolonged air exposure. The N-terminal hexenoyl group confers resistance to dipeptidyl peptidase-4 (DPP-4) cleavage. Avoid repeated freeze-thaw cycles.

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STRUCTURAL NOTES

Tesamorelin is a 44-amino-acid synthetic analog of human growth-hormone-releasing hormone (hGHRH(1-44)), modified by N-terminal acylation with a trans-3-hexenoyl group. The lipid modification at the N-terminus stabilizes the peptide against degradation by dipeptidyl peptidase-4, the primary inactivating protease of native GHRH. The C-terminus is amidated, consistent with the native processed GHRH(1-44) form.

SELECTED PEER-REVIEWED REFERENCES

1. Falutz J. et al. - *Effects of tesamorelin, a growth hormone-releasing factor, in HIV-infected patients with abdominal fat accumulation: a randomized placebo-controlled trial with a safety extension* - Journal of Acquired Immune Deficiency Syndromes, 2010 - DOI: 10.1097/QAI.0b013e3181e7f1e0
2. Stanley T.L. et al. - *Effect of tesamorelin on visceral fat and liver fat in HIV-infected patients with abdominal fat accumulation: a randomized clinical trial* - JAMA, 2014 - DOI: 10.1001/jama.2014.2129
3. Ferdinandi E.S. et al. - *Non-clinical pharmacology and safety evaluation of TH9507, a human growth hormone-releasing factor analogue* - Basic & Clinical Pharmacology & Toxicology, 2007 - DOI: 10.1111/j.1742-7843.2007.00094.x

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.